

CS598 HPC. Homework 3 - Parallel LU

Due: Friday 10/9/26

Objective: Develop fast distributed-memory LU factorization.

In these exercises, we will not consider pivoting, so your test matrices should either be SPD or diagonally dominant.

One idea for $A \in \mathbb{R}^{n \times n}$, $n = N - 1$,

$$A = S + aI, \quad a = N, \quad (1)$$

$$s_{ij} = \sqrt{2/N} \sin(\pi i j / N), \quad i, j \in \{1, \dots, N - 1\}^2. \quad (2)$$

Q1. Extend the provided serial code to support distributed-memory parallelism using a (P, Q) tiling as discussed in class, in the `lu.pdf` document, and in Chapter 4 of van de Velde.

- Verify that you are getting the correct factorization for several (P, Q) and n values.
- Explore strong- (n fixed, (P, Q) increasing) and weak-scaling ($n^2/(PQ)$ fixed as (P, Q) increases) for several values of (n/PQ) .
Please use plots—possibly log-log—to illustrate the performance behavior.
- How large must $n^2/(PQ)$ (the local memory footprint) be in order to see 80% parallel efficiency?
- What values of GFLOPS are you observing?

Q2. Extend your BLAS1 code of **Q1** to support BLAS3-based blocked factorization, with pivot blocksize $b \times b$ (e.g., for $b = 4, 8, 16, 20, \dots, 128$). Use vendor-provided `dgemm` (see below) to extract max performance.

- Answer the same questions as for **Q1**.
- Additionally, explore and report on the optimal blocksize, b , for your test cases.

Vendor-Supplied dgemm:

AMD EPYC: AOCL (AMD Optimizing CPU Libraries)

- Its BLAS is AOCL-BLIS, AMD's tuned fork of BLIS, with kernels for Zen through Zen 4/5, including AVX-512 on Zen 4 and later.
- Link with `-lblis` or `-lblis-mt` for the threaded version.

Intel Ice Lake: oneMKL (Math Kernel Library, now part of oneAPI)

- It has AVX-512 kernels for Ice Lake. It's free to use and redistribute.
- MKL also runs on AMD. Older versions steered AMD chips onto slower code paths, which newer releases have largely fixed. On EPYC, AOCL-BLIS or OpenBLAS is still the safer default.